

This unprecedented access through free services like Google SketchUp and other similar initiatives is allowing even casual users a look into what was once a specialised tool.

In the world of design, software is used in all fields; from architecture to medical sciences. The manufacturing industry is now significantly dependent on the benefits computer aided software provides and developments in design software's is as prolific as any other technological field. This innovation has come about over the last few decades to a point where it's applications in 3D printing are making it possible for even home user's a doorway into designing their own products. To better understand its importance and usage in 3D printing we shall take a walk-through a brief history of its development and assess the softwares available, both commercial and open source, in this chapter.

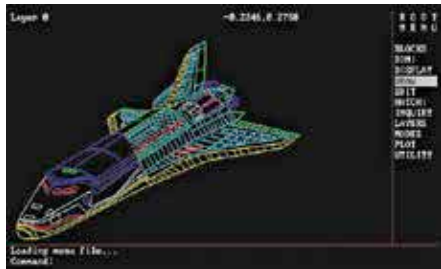
History of 3D Printing Software

As we know, computer aided software's or CAD softwares, are an intrinsic part of the 3D printing workflow. But more than that, they are the very seed from which any 3D printing project blooms. The best way to consider a CAD software is as the equivalent of word processing softwares used to print text files.

The original creation of CAD began in the late 1960s and was mainly used for creating blueprints for building design. Priced at nearly USD 500,000 by the Control Data Corporation, these software embedded machine mainframes with not only a giant piece of hardware but also a remarkable

leap over manual hand designed methods, capable of rendering a design model in less than a minute. Interestingly, today's basic cell phone processors can accomplish the same task over thirty times every second.

By the early 1980s a company called Autodesk was in the process of taking this software segment generations ahead. Their goal was to create a revolutionary software that would be faster, more powerful and most importantly - cheaper. This was the first ideation of the "word processor



An ancient relic compared to today's slick Autodesk products, MicroCad was a breakthrough of its time.